

Design FMEA VBF-T2R1SINGLEMODEL-DFMEA-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

Design FMEA (qualitative, simulation-signal based) — VBF-T2R1SINGLEMODEL-DFMEA-01

> Qualitative version, based on simulation signals; severity/occurrence three-level (high/medium/low),
> basis = magnitude of the simulation value vs threshold. Generation date 2026-08-25.

Failure mode	Failure cause	Simulation signal	Severity	Occurrence	Mitigation (design side)
Negative-electrode lithium plating (fast charge)	4C reaction-zone crowding near separator	ap_min +0.112 V / +0.111 V (margin > 0)	high (if occurred)	low	Implemented: 250 um anode + neg i0 = 2.0 A/m2; conservative 25 C start verified
Thermal runaway from overcharge	heating in 4.7 V charge phase	DFN overcharge unsolved (IDA failure at 4.7 V phase — recorded); TR coupling not triggered at 322.7 K	medium	unknown	Physical overcharge test required; BMS voltage clamp (not simulated)
Thermal runaway from nail	internal short heat	triggered, T_max 408.4 K	high	medium	External short protection / thermal fuse (design-side, not simulated)
Electrolyte oxidative decomposition	voltage window vs HOMO	not assessed (funnel_voting OFF: xtb HOMO line unavailable; real_compute=false)	medium	unknown	Conservative 4.2 V cut-off retained
Insufficient capacity / energy density	thin electrodes / low utilization	ED 358.0 vs 327.18 Wh/kg (+30.8 margin)	medium	low	100 um cathode retained; margin documented
Low-temperature performance loss	sluggish transport at 253.15 K	retention 99.59 % (>= 90 criterion)	medium	low	Verified at -20 C; below -20 C untested
SEI overgrowth	solvent diffusion through film	194.6 nm @100 / 439.7 nm @500 vs 500/550 nm	medium	low	D_ec 8e-20 m2/s densified film (bridge extension, recorded as estimate)

Conclusion

Highest-risk items: overcharge TR (virtual test unresolved — physical test mandatory) and nail TR (triggered in virtual test — protection measures required). Complete FMEA incl. process/supplier failures: N/A (beyond pure simulation boundary).